# Directional integrals of derivatives of the Fourier transform are controlled by a weighted Sobolev norm

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*Editorial note.* The counted statement is the author's labelled lemma 'gnorm' (source lines 1037-1042); the proof environment begins on the next line after \end{lemma}, so the pairing is by adjacency.

*Editorial note.* The retained context carries the two lemmas the proof invokes by label, namely the ordering-of-moments lemma and the recalled L2 trace lemma.

*Editorial note.* The retained preamble is the publisher layout adapter plus the native macro, operator and theorem block of the same pinned source.

*Editorial note.* Every label used inside the retained text is defined inside one of the retained spans, so the package declares no external reference.

*Editorial scope.* One original result of the pinned publisher article, counted once: the directional-integral bound for derivatives of the Fourier transform (author label gnorm).

*Allowed standard prerequisites (editorial).*

- Weighted Sobolev norm H^{sigma}_M. The norm is the sum over multi-indices of order at most M of the L2 norms of <nabla_k>^sigma (k^alpha g).. It is the norm appearing on the right-hand side of the statement and reordered twice inside the proof.
- L2 trace lemma. For g in H^s(R^d) with s > (d-1)/2 and a straight line C, the L2 norm of g on C is bounded by the H^s norm of g.. It is applied to the directional line at source line 1044 to pass from the one-dimensional integral to a full space integral.
- Cauchy-Schwarz and Plancherel. Standard inequalities converting an integral of |f| into an L2 norm and back between physical and frequency space.. Cauchy-Schwarz opens the proof at line 1044 and Plancherel converts back at line 1049.

*Source reference.* Original lemma 5.2: sources/hartree-phase-mixing-jep-2026-353/source0.tex, source lines 1030–1030. Original recalled L2 trace lemma invoked at source line 1044. It is retained as a labelled pointer rather than as context because its own statement carries a citation whose entry is not part of the retained publisher bibliography; the counted lemma's proof is retained in

full and nothing is replaced by this reference. The reference number was inferred by counting the theorem-class environments of that section of the pinned source; it is not confirmed by the retained bibliography or by a label-number dump of the original, so it must be read as a navigation pointer only.

**Lemma 0.1** (Ordering of moments)**.** *Let $\sigma \geq 0$, let $M \in \mathbb{N}$, and $g \in H^{\sigma}_{M}(\mathbb{R}^d)$. Then the following norms are equivalent:*

$$\|g\|_{H^{\sigma}_{M}} := \sum_{|\alpha|\leq M} \|\langle \nabla_k \rangle^{\sigma}(k^{\alpha} g)\|_{L^2} \sim \sum_{|\alpha|\leq M} \|k^{\alpha}\langle \nabla_k \rangle^{\sigma} g\|_{L^2}.$$

**Lemma 0.2.** *Let $n \geq 0$, and let $\alpha$ be a multi-index. Then, for every $k \neq 0$,*

$$\int_0^{\infty} \langle s\rangle^n \left|\partial_k^{\alpha}\widehat{g}\left(\tfrac{k}{|k|}s\right)\right| ds \lesssim_{d,\delta_0,n,|\alpha|} \|g\|_{H^{n+1/2+\delta_0}_{|\alpha|+\lceil d/2\rceil}}.$$

*Proof.* Applying Cauchy–Schwarz followed by Lemma 5.2,

$$\begin{aligned}\int_0^{\infty} \langle s\rangle^n \left|\partial_k^{\alpha}\widehat{g}\left(\tfrac{k}{|k|}s\right)\right| ds &\lesssim_{\delta_0} \left(\int_0^{\infty} \langle s\rangle^{2n+1+2\delta_0} \left|\partial_k^{\alpha}\widehat{g}\left(\tfrac{k}{|k|}s\right)\right|^2 ds\right)^{1/2} \\ &\lesssim_d \left(\int_{\mathbb{R}^d_k} \left|\langle \nabla_k\rangle^{\lceil d/2\rceil}\left(\langle k\rangle^{n+1/2+\delta_0}\partial_k^{\alpha}\widehat{g}(k)\right)\right|^2 dk\right)^{1/2}.\end{aligned}$$

Then, by Plancherel and Lemma 0.1:

$$\begin{aligned}\int_0^{\infty} \langle s\rangle^n \left|\partial_k^{\alpha}\widehat{g}\left(\tfrac{k}{|k|}s\right)\right| ds &\lesssim_d \left\|\langle x\rangle^{\lceil d/2\rceil}\langle \nabla_x\rangle^{n+1/2+\delta_0}\left(x^{\alpha}g(x)\right)\right\|_{L^2} \\ &\lesssim \sum_{|\beta|\leq\lceil d/2\rceil} \left\|x^{\beta}\langle \nabla_x\rangle^{n+1/2+\delta_0}\left(x^{\alpha}g(x)\right)\right\|_{L^2} \\ &\lesssim_n \sum_{|\beta|\leq\lceil d/2\rceil} \left\|\langle \nabla_x\rangle^{n+1/2+\delta_0}\left(x^{\beta+\alpha}g(x)\right)\right\|_{L^2} \\ &\lesssim_{\alpha} \|g\|_{H^{n+1/2+\delta_0}_{|\alpha|+\lceil d/2\rceil}}.\end{aligned}$$

□

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